

A
METHOD
OF
TANNING
WITHOUT
BARK.



DUBLIN:
Printed by A. RHAMES, MDCCCXIX

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*** START OF THE PROJECT GUTENBERG EBOOK A METHOD OF
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Transcriber's Note: This old text is preserved as printed apart from the correction of the following apparent printing errors:

[Page 13](#) "Extroordinary" changed to "Extraordinary"

[Page 20](#) "luberous" changed to "tuberous"

[Page 26](#) “perpendicular” changed to “perpendicular”

[Page 34](#) “Veiws” changed to “Views”

[Page 35](#) “cheifest” changed to “chiefest”

Text printed in an “Irish” typeface is denoted by use of the font ‘Old English Text MT’ or ‘Old English’. Without one of these fonts installed you may not be able to discern any difference from the surrounding text.

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TO
HIS EXCELLENCY
William Conolly, Esq;
SPEAKER
OF THE
House of Commons of *Ireland*.

And to the Rest of the
Members of That HOUSE.

THIS
METHOD
OF
TANNING without BARK

Is humbly inscribed

By Their most

Obedient Servant,

William Maple.

THE
Method of Tanning

WITHOUT BARK.



In Political Computations, the following Principles are universally allowed.

That the Riches of every Country arise from the Natural Produce of the Earth; and the Labour and Art, imployed in improving the same.

That the Quantity of Native Commodities, the Source of Riches, is in Proportion to the Number of People, fully and usefully employed.

That every industrious Person adds to the Wealth, and Support of a Nation; and every Idler, detracts from both, by living on the Labour of Others.

That it is the best Policy in any Government, to encourage, and provide for the full Imployment of the People; since thereby, the Number, Wealth, Strength, and Quiet of the Inhabitants are increased.

That the Gain, or Loss of any Nation by Trade, is known, and determined by the Proportion, which the Exports bear in Value to the Imports. On this depends the absolute Quantity of Money in a Nation; the Relative Quantity of the several Species, that composeth this Money, being ascertained by the Value imposed on those Species; either as it agrees with, or differs from the Value set thereon by other Nations.

A Nation certainly loses by a Trade, whose Exports are not compleatly manufactured, and whole Imports are fully wrought.

A few Manufactures will serve to exchange for great Quantities of the simple Produce of the Earth. The Value of Manufactures, being principally owing to Labour and Art.

If upon these Principles, we examine the State of the Trade of *Ireland*; I fear we shall find it, in a very declining, and ruinous Condition.

Our Exportations are made up of Wool, Woorsted, Linnen, Yarn, Beef, Butter, Cheese, Tallow, Hydes, Skins, and a few other Articles.

We are not allowed to export our Wool and Woorsted, but to a few Ports in *England*, and are prohibited from sending thither, or any where else, any other of our Woollen Manufacture. Linnen, and some tanned Hides are the only Commodities we export, compleatly Manufactured; Yarn and Woorsted may be accounted but as half-wrought; Almost all the other Articles are but little removed from the State, Nature has given them to us, and have little or no Labour or Art employed about them.

At the same Time, we export so liberally the Growth of our Country, and so little of our Manufactures, we may observe all our Importations are of such Commodities, as are not capable of farther Improvement; such as *East-Indian* and *West-Indian* Goods, Druggs, Wines, Spirits, Manufactures of Wool, Silk, Linnen, Iron, Brass, Glass, &c. Nay, the common Produce of almost all Countries, as Corn, Timber, and Fireing, are imported in great Quantities to supply our Necessities.

By this Management of our Trade, we maintain many Thousands in foreign Countries, and drive Numbers of our own People from us, or leave them to starve at home, for Want of Employment.

If to this View, we add that mighty Drain of our Bullion, made by those Gentlemen, who, having great Estates in this Kingdom, choose to spend them any where, rather than at home: We can not be at a Loss, how to account for the Ballance of Trade, running against us, for so many Years past.

The most certain Way left to retrieve our Affairs, is to lessen our Imports, and to encrease our Exports, especially of Manufactures.

The Commissioners of the Linnen Manufacture, by an unwearied Application and uncommon Skill, have contributed greatly to the Value of our Exports, by the Advancement of the Linnen Manufactory. And I don't doubt but that in several other Articles, divers Expedients and Means, may be thought of, for promoting a greater Consumption of our Own, and lessening that of foreign Commodities; and as one Instance, I shall beg Leave to propose an Improvement in our Tannage, by which I apprehend, we shall be enabled to Tan with the Produce of our own Country all the Hydes that now are Exported Raw; and thereby increase the Value of our Exports, employ many Thousands of our People, and avoid the Necessity of Importing Bark.

To set this Matter in a clear Light, I shall lay down, in the first Place, the Export of Green Hydes, and the Import of Bark, in the seven following Years, for the Whole Kingdom, as the same has been extracted out of the Custom-House Books.

<i>Years.</i>	<i>Gr. Hydes.</i>	<i>Bark B's.</i>
1721	46,847	46,556
1722	86,004	45,794½
1723	123,514	47,824½
1724	103,477	60,740½
1725	69,858	48,407
1726	60,441	45,279
1727	57,373	50,598

In the next Place it is to be observ'd,

That Bark, according to its Goodness, is sold from 6 to 10, and in some Years, 11 Shillings, the Barrel.

That Cow-Hydes weigh from 56, to 84 Pounds each; Bullock Hydes, weigh from 84 to 140 Pounds each, and both are sold from 15 to 21 Shillings, the hundred Weight.

That Soles, are generally sold from 8 to 8 Pence half-penny the Pound.

That Hydes generally lose about half their Weight, in Tanning.

From these Principles, when thrown into mean Proportionate Quantities, it is evident:

That the Quantity of Bark, annually imported, is 50,000 Barrels, and the Value 21,000 Pounds.

That the Number of Green Hydes, annually exported, are 80,000, and their Value 57,000 Pounds.

That if these Green Hydes were tanned before Exportation, their Value would be 123,000 Pounds.

That the Difference, arising from the Different State of these Hydes, on Exportation, is 66,000 Pounds, which Sum added to the 21,000 Pounds, the Value of the imported Bark, making in the whole 87,000 Pounds, must be

accounted, as so much gained to the Kingdom, should our Tannage be effected by our own Produce and Labour.

The Nations, that demand this Export of those Green Hydes, are those, whose black Cattle are not bred, in such Numbers, as to supply their indispensable Demands; Nor will their dry Hydes, which they receive from their Settlements abroad, admit a thorough Tannage; They, in hopes of putting their Neighbours, who have a Redundancy of Hydes, under a Necessity, of parting with them unmanufactured, have prohibited, by the severest Penalties, the sending us any Bark. While we, on the other Part, have not only complied with this detrimental Export; but also have been so regardless of our own Interest, as to permit several Tanners, some from this City, to go, and instruct them in an Art, they almost were Strangers to.

The *English*, from a true Sense of the Disadvantage of such a Trade, have subjected the Exporter of a Green Hyde, to the Penalties of 500 Pounds, and a Disability to trade in Leather, ever after. *Ann.* 14. *Car.* II.

Such an Act, even in our present Circumstances, may, by some, be thought adviseable for us; Because, such Nations, as now gladly receive our Green Hydes, finding a Stop, put to their usual Supplies, would be under a Necessity of receiving our tanned Hydes, and selling us their Bark. Leather being a Commodity, that does not depend on the Fancy; but, is absolutely requisite, not only to the Luxuries, but also, to the Necessities of Mankind.

At present, Our Tanners find a great Scarcity of Bark, except those residing in some considerable Sea-Port-Town, and are under just Apprehensions of a greater. Several in the Country have been obliged to lay aside their Pits, and take to other Employments; the usual Supplies failing. Our Neighbours who will permit any Bark, to be sent us, are in a Manner exhausted. Our *American* Plantations are too remote to find their Account, in sending any great Quantity. The Produce of our Country is scarce a twentieth Part of what is used in Tanning. This renders our Tannage dear, and often imperfect, so as not to serve a foreign Market, to Advantage, and Reputation.

But if we had Materials, of our own Growth, easy of Acquisition, and Propagation, that would tan, in every Respect, as well as Bark; were these Materials, as to their Powers, well ascertain'd, and confirm'd by a general

Experience, the making an Act, to prohibit the Exportation of Green Hydes, would be judged by all to be Adviseable, and Useful to the Kingdom.

When the apparent Qualities of Bark, are considered; and that by it's insinuating into the Pores, and incorporating with the Substance of the Hyde, it produces a Firmness, Strength, and Consolidation of the Parts; It is very surprizing, that other Materials, of the same Apparent Qualities, have so long remained neglected, and unapplied to the same Uses; when the Consequence is of such Moment.

Tormentil, and *Cinquefoil* Roots, are ranged in the same Medical Class; have the same apparent Qualities, especially the first, in a more eminent, and evident Degree, than Bark, it self.

On Tryal, they have answered beyond Expectation: The *Cinquefoil* did not give so good a Colour, as the *Tormentil*, which in all Respects, as to Colour, Bloom, Substance, Solidity, and Weight, in the Tanned Hyde, compleatly answered, and in much less Time, than when Bark, (even of the best Kind, or Growth) is used.

Tho', what is here asserted, is grounded on several Years Experience, and confirmed by the Opinion and Testimony of Mr. *Philip Cooley*, Tanner, who from the Beginning hath been concerned in the several Tryals, and to whose Skill and Industry a Part of the Subsequent is owing. Yet to give the utmost Satisfaction, and as I think my self in the strongest Manner bound by the Encouragement, hitherto given me, by the *House of Commons*, to render the best Information I am able, in an Affair of some Consequence to the Public, I shall give a most faithful Account of the Progress that has been made, and the Success that hath attended this Way of Tanning with relation to the Goodness of the Leather, the Time, and Charge of Tanning, and other Circumstances; and shall add several Observations, with Directions for finding, and propagating these Roots, in great Plenty, that all Persons may be encouraged to carry this Method of Tanning, into a general Practice, which as it will be for their private Advantage, so it will not fail to promote the Public Benefit.

In Order to this, I shall in the first Place beg Leave to place this Affair in the Light it appeared in before the *Committee* of the *House of Commons of Ireland*, to whom my Petition was referred by the *House*, in which Petition it was alledged, that I had established a Method of Tanning, without the Use

of any Barks, to as great Perfection, cheaper, and in less Time, than with Bark, by Vegetables of our Natural Produce, and of easy Culture.

The *Committee*, were pleased, for their greater Satisfaction, to summon a Great Number of Tanners, Curriers, Shoemakers, and others, to attend, to give their Testimony in relation to the several Specimens, and Proofs, that were to be laid before them.

In respect to the Goodness of the Leather, so tanned, the following Specimens were exhibited.

Nº. 1. A Calve's Skin from the Ooze.

Nº. 2. A Calve's Skin, uncurried.

Nº. 3. A Calve's Skin, curried: each of these, when tann'd and dry'd, weighing about three Pounds.

Nº. 4. A Calve's Skin, very strong and large, when tanned and dried, weighing about Six Pounds.

Nº. 5. Seven or Eight Pair of Soles, of a Calve's Skin, rais'd.

Nº. 6. A Bend of a Bullock's Hyde for Harness.

These were all proved by *Thomas Cooley* and *Patrick Shale* to be tanned without Bark, and with the Roots.

Mr. *Henderson*, Master of the Tanners, said he was diffident of the Goodness of the Specimens, that he should have been thoroughly satisfied: had he known, how the Leathers had proved, if worn, by some Porters, or Chairmen. That the Colour of Number 2, or 3, was not so good as some tanned with Bark, producing a Calve's Skin curried, to compare with them, but did own, all the Specimens were thoroughly struck, and well tanned: That if they were in a Shop, intermixt with Skins, tanned by Bark, he could not distinguish them; and that, what tanned Numb. 5 and 6, would tann, Time and Quantity allowed, the thickest Bullock's Hyde.

Mr. *Dobson*, Tanner, affirmed the Specimens were not well tanned, because they did not come up to the Colour of the Skin produced by Mr. *Henderson*, but in other Respects, the Leathers appeared, to be as good in their respective Kinds, as any tanned with Bark.

Mr. *Nicholas Gibson*, Master of the Curriers, affirmed that the Skin produced by Mr. *Henderson*, was of an Extraordinary Colour; That there were not ten such in *Dublin*; That the Difference in Colour was but very small, and in regard even to Colour, would choose those tanned by the Vegetables, as soon as Mr. *Henderson's* Skin; That all the Specimens appeared to be very well tanned, and that he could in no wise distinguish them from Bark-Tannage.

Mr. *Devereux*, Currier, being sent out, returned, producing two Calves Skins tanned by Bark, which he took indifferently, and without choosing, from the next Currier's Shop. They were allowed by all, to be of a general or common Colour; and being compared were much exceeded by the Specimens, and particularly, as to Colour.

Mr. *Edmund Sheild*, Currier, said he never saw better tanned Leather, than the Specimens, that their Colour was very good and, that what tanned Numb. 5. would tann any Hyde, whatsoever.

Mr. *Brookfield*, Tanner, reported the Specimens exhibited, were well tanned and thoroughly struck; that neither He, or any other could distinguish them, from Bark Tannage; and that what tanned Numb. 5, could not fail in any Tannage.

Mr. *Veckers*, Tanner, allowed the Specimens to be very well tanned; and gave it, as his Opinion, that what tanned Numb. 5 and 6, would tann the thickest Sole Leather.

Mr. *John Gades*, Shoemaker, after giving his Testimony, that all the Specimens appeared to be exceedingly well tanned, produced a Shoe, made by him, of the Leather of this Tannage, affirming the Leather, wrought very well, and that the Soles exhibited, under the Hammer, all the Signs of extraordinary Leather.

Mr. *Ellis*, Shoemaker, reported the Specimens were very well tanned: That, Leather either dead, or highly tanned, would never carry so beautiful a Colour as Leather otherwise tanned. That, the Sole Leather appeared to be very well tanned; and that in his Opinion this Tannage, would in every respect, answer the same End, or Intentions, as Bark Tannage.

Mr. *James King*, Shoemaker, said all the Specimens, were effectually tanned.

Mr. *Richard Norris*, Shoemaker, produced a Sole of this Tannage, beaten; asserting he never saw so good a Sole of rais'd Calve's Skin, that it beat firm, solid, and without spreading, or Fuzziness: which are the distinguishing Characters of good Sole Leather, he desired it might be cut in several Places, that the Closeness, and Firmness might be view'd, which accordingly was done, and answering Expectation was approved.

There were a great many other Tanners, Curriers, and Shoemakers, who were of the same Opinion. But considering, their Testimonies, would be but so many Repititions; I shall not recite them.

Dr. *Richard Helsham*, Professor of Natural Philosophy in *Trinity College*, said I had communicated to him, what was used in this Tannage, which appeared to be well adapted to tanning, and would not fail to give a sufficient Strength, and Rigidity to the Fibres. Adding he wore a pair of Shoes, the Upper Leathers of this Tannage, daily for four Months, and being willing to try the utmost of the Leather, had new Soles put to them; that since he had worn them about a Month: showing them, without any Cracks, or Defect.

Patrick Carrol, produced two Slips of curried Calf Skin; the one tanned by Bark, the other by the Roots, each three tenths of an Inch broad, and ten Inches long, the first was the thickest, and weighed, a tenth Part more than the other. He said, he was present with several others, when it broke, by a Weight of 129 Pounds being hung to it; That the Slip of this new Tannage, being tryed with the same Circumstances, bore 29 Pounds more than the 129 Pounds, before it either yielded, or broke.

In respect to Time.

That the Specimens, N^o. 1, 2 and 3 were tanned in five Weeks, N^o. 4 in nine Weeks, N^o. 5 in four Months, without any Hurry, extraordinary, or illegal Methods: was proved by Mr. *Thomas Cooley*, and *Patrick Shale*; on the other Side, it appeared, that in tanning with Bark, from three to five Months were required for Calves Skins, from seven to nine were requisite for Soles of rais'd Calves Skins, and from ten to thirteen Months for Bullocks Hydes, which by this Tannage, may be effected in nine or ten.

In regard to Cheapness.

Mr. *Philip Cooley*, affirmed that six Pounds of the Dry Roots, added to some old expended Ooze, tanned Six Calves Skins, three of which were produced under the N^o. 1, 2, and 3; and that he believed sixteen Pounds would have done as much from the first, or without any old Ooze.

In relation to the Cheapness.

Part of the Roots, I have used, I purchased at three Farthings, the Pound; but of late Years, and being willing to engage the Privacy of the Gatherer; I have given a Penny; they lose about a third in Drying: Allowing these as Facts and taking it at the dearest, less than five Shillings worth of Roots, will go farther, than a Barrel of the best Bark.

As to their Plenty and Propagation, Dr. *Helsham* testified, they grew in great Plenty as he had observed, in several Counties of this Kingdom, and were very easily propagated.

Dr. *William Stephens*, Botanic Professor in *Trinity College, Dublin*, in his Examination declared, that they grew in great Plenty about *Dublin*, and in several Counties he had observed great Quantities, chiefly on Mountains, and barren Places: That, they were easily to be propagated, in almost any kind of Soil, and that he had some flourishing in the Physick Garden, belonging to the University.

The Resolutions, that the House entred into *January 10*, might very well have sufficed to have recommended a General Practice. But to remove some Prejudices and to obviate some false Representations, I have impartially recited the several Facts and Testimonies, on which they were founded.

That all People may easily know and find out these Plants, I shall here give a Description; which though tedious, I hope will be excused.

Cinquefoil, in *Latin*, *Pentaphyllum vulgare*, in *Irish*, Tuigbeag (*Tuigbeag*) Tuigvear Muirre (*Tuigvear Muirre*) and in some Counties, Meagigh, (*Meagigh*) is a Plant, whose Roots are very long, about a Finger's Thickness, of a dark brown or reddish Colour, and very astringent in Taste; the Leaves are born at the Extremity of a small Stem, five together, deeply indented on their Edges, of an oblong Form and hairy; on the Grand Stem, at the Nodes, shoot forth several smaller, terminating with a small Yellow

Flower, composed of six small Leaves surrounding a Number of Threads, from whence ariseth a Cluster of Seeds.

It flourishes in every Hedge, sandy, loose and deep Soil: the Roots which are to be preferred, are the thick, long, and of the darkest Colour.

Tormentil, or *Septfoil*, in *West Meath*, called *Neauhnadis* (*Neauhnadis*) in *Ulster*, *Menedin* (*Menedin*) or *Neauhnid* (*Neauhnid*) in *Connaught*, *Levenet* (*Levenet*) in the County of *Mayo*, *Fenede na Muc* (*Fenede na Muc*) this Plant, hath a thick, tuberous Root, of a reddish brown Colour, sometimes covered with a black Skin, of a very astringent Taste, and shooting out many Fibres; the Leaves are generally seven, set on the Extremity of a Foot Stalk, of a dark Green Colour, deeply indented on the Edges and hairy, from amidst them ariseth several small weak Stalks, of a reddish Colour, and hairy, about 12 or 18 Inches in Height, knotted; at every Knot, or Joint, is a Foliage, and two or three Foot Stalks, some bearing Leaves, and one longer than the rest the Flowers consisting of four Leaves, of a light Yellow Colour, to which succeeds the Seeds; the flower Cup, hath Eight Leaves, four large and four small, the Root when Vigorous sends forth several Runners, the Stalks of which are of a reddish Colour, and hairy that spread over the Ground; at every Joynt, from one Part ariseth two or three Stems, each bearing at the Extremity five Leaves, from the other Part, descend several Fibres that become Roots.

It grows almost every where, but chiefly delights in mountainy, barren and shallow Soils; the Roots seldom striking four or five Inches deep. It hath been observed by Dr. *Stephens*, to grow all along the Banks of the *Doder*, from *Donore-Brook* to *Old Baun*, upon Roch Town Hills; it grows on the Borders of a great Bog for several Miles, by *Rathmullian* in the County of *Meath*. In the County of *Wicklow*, Thousands of Cart-Loads might be with Ease collected; Nor do I know a Mountain or Bog without Plenty of it.

In the Figure, (*a*) denotes the Root, (*b*) the Flower in the Bud, (*c*) the Flower expanded, (*d*) the Knob of Seeds, (*e*) the Runners, (*f*) the Flower Cup. I have exhibited but few of the Leaves, Stems, and Runners to avoid Confusion.

It is observable, that from the Bark of the Stalk of this Plant, at its breaking forth from the Root, excrementitious Tubercles or Knots,

resembling Oak-Galls have been often discovered, and like them are subservient to the Propagation of Insects; which, considering the extream nice, and distinguishing Taste of those Animalculas, may well be allowed as a Proof, that their Juices are similar.

The Roots growing on Mountains are small, strong, and often have a black Coat; those that grow on the Sides of Loughs, and in Bogs, are large, of a lighter Colour; some I have had of the Latter, that have weighed each above half a Pound.



When you chose the Roots, prefer the large, knotty Roots, that when broken exhibit a blood red Tincture intermixt with the Brown, and those that when cut, leaves on the Knife a strong purple Colour.

The best Time for collecting these Roots is in the Winter Season, or early in the Spring, and tho' the Seed-bearing Stalks, and Foliage are decayed, yet there are several Foot Leaves, that remain and direct the Inquirer.

And tho' these Plants grow wild all over the Country, in such Plenty, as to be more than sufficient to tann all the Hydes, and Skins of the Kingdom; and little Apprehension of a Deficiency, for in digging up the Roots, there will be left some broken Parts, some Fibres, which will Vegetate and give a Supply. A common Root, on a moderate Computation will produce 8 or 10 hundred Seeds, these shedding, and dispersed by Winds, and other Accidents, some will take Root, and carry on a Succession.

And in regard, it may be objected, that the Trouble and Expence, in gathering them, when in so dispers'd a Manner, would be very great; and a constant Supply uncertain: And because the Roots, if raised in such Soils, as shall be found to agree with them, would thereby become larger, stronger, as well as cheaper, I have endeavoured the Culture, especially of the *Tormentil*, and with such Success, as to have from each square Yard three Pounds weight of the Roots.

An Acre contains 7,840 square Yards; but, for Paths, Wastes, and inclosing, I would allow 1,800 Yards, not to be employed; As the first Year they will be little more than fibrous, I would allow three Years for their Growth, tho', if another Year was added, the Crop by an Increase would sufficiently reward, not only as to Quantity but Quality; as the best of Lands are not required; I would state the Expence of that Culture as follows,

Three Years Rent at 5s. the Acre,	0	15	0
Plowing twice and Harrowing,	0	12	6
Planting,	0	7	6
Houghing,	1		
Digging up the Roots,	1		
Total	3	15	0

Suppose each square Yard, should produce but the Moiety of what before was mentioned, and sold but at three farthings the Pound, the Amount will be 27*l.* 3*s.* 6*d.* from whence a Profit ariseth beyond most of our Cultures, now in general Use, and the Tannage may be accomplished, at about a Moiety of the present Expence.

Probably, this Estimate of the Charge and Profit, may not be so exact as to be strenuously defended; but if defective, there is Room enough for any

reasonable Addition or Deduction, without much Prejudice to the Argument; different Soils, often one and the same will be attended with different Charges; The finer, the Earth is made, or the more it is broken, before Planting the better Vegetation succeeds; at first shooting up of the Herbage, particularly where the Ground is foul, houghing, or pulling up the large Weeds, will be necessary, that the Growth of the *Tormentil* may not be chequed; in a little Time it will clear its self. As the Roots run superficially, not streeking at most 5 or 6 Inches in perpendicular Depth, the Charge of digging them up cannot be so much, as that which attends Potatoes.

The Propagation may be effected by planting the Crowns of the Roots, (that Part, from whence the Herbage ariseth) in either Autumn or Spring, in the same manner as Horse Raddish, splitting the Crown, so as to leave a Bud on every Slice, then setting them in the Ground, about an Inch in Depth and allowing each Set, 5 or 6 Inches square of Ground. When you take up a Plantation, you will have an Abundance of small Roots that arose from the shedding of the Seed, and from the Runners, that will scarce be worth the Drying, but very proper for Planting: If you cut from either Extremity of the Root, a Piece of about half an Inch in Length, it will grow very well. The whole Root, if cut into such Pieces, may be planted. In all these I believe the Charge of Planting might be saved; if you scatter them before Harrow, the passing of that over them, will cover them with Earth sufficient; or at least, the most Part of them.

You may Plant the Runners, as you do Strawberries, in the mention'd Seasons and Distances, and considering the Numbers of fibrous Roots, already shot forth, it may be allowed the speediest Method.

The Seeds ripen about *August* or *September*, when ripe, they are of brownish Colour; as they are very apt to shed, the Time of ripening must be heedfully regarded; It would be adviseable to cut the Hawm or Herbage, a little time before the Seeds ripen, laying it on a Cloth, or winnowing Sheet in the Sun, when dry, rub the Seeds out with your Hands, and separate them from the Chaff. In *March* the Ground being prepared, you may sow and harrow, or rake them into the Ground.

The Cutting of the young Stems, if planted in *May*, will grow.

The Roots when collected, are to be washed very clean, from the Earth that will remain with them, for otherwise, that Dirt when in the Layers,

settlesh into the Pores of the Hyde or Skin, and causeth Numbers of Spots, and Discolourations.

In fair Weather, it not being a succulent Root, they will sufficiently dry if spread on a Floor, or a clean Part of the Field; a little Wet, will in no case damage them. In case of very wet Weather, and continued, they may be Kiln dried, in the same Manner as Bark.

When dried, they must be bruised or ground, either more or less, according to the Uses design'd, or to the Proportion you require them to spend in. In case they should not be sufficiently washt, when they are pounded, or ground a little, it would be adviseable to sift the Dirt, or Earth, which by this Means is broken, and seperated from them.

When thus prepared, you must proceed in the same Manner, and lay away, intirely, with the same Circumstances as with Bark.

It hath often been objected, That if this Method should obtain, Planting would be discouraged, and so prejudice the landed Interest. That it is used in the *Canary Islands* and that though it may be done effectually there, yet the Produce of our Country might have Qualities very different, or not exalted to such a Degree, as to be of a requisite Force.

To these, I would answer.

That Bark would always bear a Price, in Proportion to the Rate of what is substituted in lieu thereof; It would be far from being rendered useless, but only suffer a Diminution in Value.

That Diminution would be more than ballanced by the accelerated Growth, and Worth of the Timber. At present we fall our Woods so young, and at such improper Seasons, on Account of the Bark, that the Timber is neither so good, and little more than fit for Ribberies, and Cabbins; whereas otherwise, was the present Inducement, the great Price of Bark diminished, we should have full, and well grown Timber, cut at a due Season, and proper for Edifices, and Shipping.

The Improvement of those Lands that now are unprofitable, or not far from such, would be another Ballance, if not wholly, yet in Part, to the apprehended Injury.

The Use of this Root hath not been confined to the *Canaries*. The Reverend Mr. *Lucas Jacobson Debes*, in his History of the Islands of *Feroe* says Page 120, “Here groweth a huge Quantity of *Tormentil*, which the Inhabitants, having no Bark of Trees to tan their Hydes with, make Use of; God and Nature having revealed to them the Dryness of that Herb, so that they tan their Skins with it, and therefore call it Bark.” And Page 271. “The Men wear Shoes of Neats Leather, tanned with the Roots of *Tormentil*.”

The first of these Islands, is in the Latitude of 28, the other in 62. Yet we find the same Roots used for the same Purposes, though probably they may differ in Degrees of Power.

A great Degree of Heat is not a Quality requisite to the Perfection of all Vegetables. *Saffron* in *France* exceeds the Produce of *Spain*, as much as the little we have in *Ireland* excells the Product of *England*.

Our Temperate Climate seems in some Degree, to be more adapted to Vegetables of this Species of Power, than those that are deemed hot. Our Bark is allowed to excel the Bark of those warmer Climates, as also our Timber.

It is a contested Point, whether the *Tormentil* is not the Plant, the *Greeks* called *Britanica*; probably so named from its plentiful Growth in that Island.

One great Advantage, arising from this Tannage should it become general, will be, that it will give Employment to a great many Poor People, that will be set to work in gathering, or raising such a Quantity of those Roots, as will answer the present annual Import of Bark; as also the Tannage of some, if not the Whole of those Hydes, we now export unwrought; besides the great Numbers, that will be employed in the encreased Tannage.

Thus, with as much Brevity, as the Nature of the Subject would admit of, I have given a Narrative, of what I hope will be as satisfactory, on a candid Perusal as I am assured, it will be beneficial to the Publick, in all its Consequences; if the Method laid down be but pursued with common Care, and Assiduity. It requiring no greater Skill than what at present is used in Bark Tannage; no Alterations in Forms, or Utensils.

If the Instructions, laid down, should appear to Any insufficient, or imperfect; I shall always with Pleasure be ready to render them, if in my Power otherwise. I have made a Plantation, very near this City, that if the Curiosity of Any should prompt them to a farther Information, in that Particular, they there may meet with it. As this Tannage is constantly carried on; it is open to the View of Any that desire it. I have several Thousands of Pounds weight of these Roots by me; I never refusing the buying of any brought to me. If any Tanner wants, he may have it at the Price, I gave to the Countryman, or be directed where, or how to be supply'd, until a Demand makes it a Market Commodity.

Our Necessities, and those daily encreasing, first prompted to this Essay. The Authority of Books, I have not relied on, but a Series of Experiments that employed several Years in the Pursuit. If I have pointed out, or cleared from Incumbrances, an advantageous Path that may lessen the Import of Bark, extend our Tannage by our own Produce, and give a Support to any Number of Poor; I regard it in no other Sense, than having done Part of that Duty, I owe to a Country, which in Gratitude, I would call mine.

I might justly incur a severe Censure, if I should conclude, without testifying with all due Acknowledgements, a grateful Sense of my Obligations, to the Honourable *House of Commons*, for the Vote they were pleased to pass in my Favour, on the Report of the *Committee*; that I had, after a full Enquiry, answered the Allegations of my Petition; and for the Resolution, they entred into, of giving all proper Encouragement to so useful a Proposal.

Far from being insensible, that all new Attempts of this Kind, are apt to subject the Author of them, to the Name of a Projector, and render him the Object of Reproach and Suspicion; when I was fully satisfied in the Facts, on which this Method is grounded; I communicated it to some Persons, who are as little liable to be impos'd on themselves, as they are above all Suspicion of imposing on others; under the Sanction of whole Characters, such an Imputation would at once be removed. By their Directions I preferred my Petition, and notwithstanding the Success was not altogether such as they desired; I hope they will not deny me the indulging a sincere Gratitude, in particularly owning the Obligations to them, who so warmly espoused my Interest, grounded upon that of the Nation. I shall venture to say no more, than as I flatter my self they can clear me from any

Imputations of Sinister or Mercenary Views; So I can promise my Endeavours shall always tend to hinder a Blush, for the Kind Representations, they have been pleased to make in my Favour; and that I shall always study to approve my self what is my chiefest Ambition to be, and what only intitles me to their Patronage; a Pretender to no other merit, but Plain dealing.

I should have published this sooner, but that I waited the Success of some repeated, as well as new Experiments; all which on the best Examination I am capable of making, have tended to an intire Confirmation; That the *Tormentil* Roots Tan all Hydes and Skins as effectually, cheaper, and in less Time than Bark, and is easily propagated.

FINIS.

*** END OF THE PROJECT GUTENBERG EBOOK A METHOD OF
TANNING WITHOUT BARK ***

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